Sistema de Controle de Impressão

Silkscreen em Máquinas de Raio-X

Requirements Document — v1.0

1. Project Overview

This application is a desktop system for managing silkscreen printing jobs on X-ray machines. It allows the operator to register machines, track how many prints have been completed, and monitor whether each machine is finished or still in progress.

A future mobile version is also planned, which will share the same database and business logic.

2. Technology Stack

Layer	Technology
Language	Java
Database	PostgreSQL
DB Connection	JDBC (via Maven)
Desktop UI (Phase 2)	JavaFX
Future Mobile	Kotlin (Android)
Build Tool	Maven
Version Control	Git + GitHub

3. Machine Entity (Maquina)

Each machine registered in the system must have the following attributes:

Field	**Type**	**Description**
id	Integer	Auto-generated unique identifier
modelo	String	Machine model name
cliente	String	Client company name
logo	String	Logo/design used for printing
totalImpressoes	Integer	Total prints expected for this machine
impressoesRealizadas	Integer	Prints completed so far

status	Boolean	false = In Progress, true = Finished
dataCriacao	Date	Date the machine was registered
dataFinalizacao	Date	Date all prints were completed
descricao	String	Optional notes or observations

4. Functional Requirements

4.1 Machine Management (CRUD)

- Create a new machine with all required fields
- Update any field of an existing machine
- Delete a machine from the system
- View a list of all registered machines

4.2 Print Tracking

- Register the number of prints completed for a machine
- Each machine has a fixed total number of prints (defined at creation)
- Status is automatically updated to Finished when impressoesRealizadas >= totalImpressoes

4.3 Status & Kanban View

- Machines are displayed in two columns:
- Em Andamento (In Progress) — prints not yet complete
- Finalizada (Finished) — all prints done
- Status updates automatically when prints are registered

4.4 Reports

- Total number of machines registered
- Total number of machines finished
- Total number of machines in progress
- List of machines with their creation and completion dates
- Filter machines by date range

5. Non-Functional Requirements

- Desktop application (Phase 1) — no internet connection required

- PostgreSQL database hosted locally
- Data must persist between sessions
- Simple and intuitive interface for daily use
- Mobile app (Phase 2) — Android, built in Kotlin, shares same database

6. Development Phases

Phase	Name	Description
1	**Backend (Console)**	Java models, PostgreSQL connection, CRUD logic via terminal
2	**Desktop UI**	JavaFX interface with Kanban view and reports
3	**Mobile App**	Android app in Kotlin connected to the same database

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